

KHLOOD OSAMA MOHAMED

Information Technology Graduate | Coding Instructor (DECI & DEMI) | Cybersecurity & Networking Specialist

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PROFESSIONAL SUMMARY

Information Technology graduate with a strong foundation in Networking, Cybersecurity, IoT, and Cloud Computing, and hands-on experience delivering technical education. Completed the DEPI Cybersecurity track as a Vulnerability Analyst & Penetration Tester and currently works as a Coding Instructor within Egypt's national digital education initiatives (DECI & DEMI). Skilled at bridging enterprise networking, embedded systems, and mobile application development into complete, real-world solutions, with an Excellent-grade graduation project on IoT-based Smart Village systems. Passionate about curriculum development and building practical, industry-aligned technical training.

PROFESSIONAL EXPERIENCE

Coding Instructor (DECI) / Assistant Coding Instructor (DEMI)

June 2026 – Present

iSchool – Digital Egypt Cubs Initiative (DECI) & Digital Egypt Marvels Initiative (DEMI)

- Deliver programming and technology courses for school students through Egypt's national digital education initiatives, using project-based learning methodologies
- Design and supervise hands-on coding labs and classroom exercises, guiding students in programming logic and computational thinking
- Provide continuous academic feedback, support debugging and troubleshooting, and collaborate with the instructional team to improve course delivery

Vulnerability Analyst & Penetration Tester

November 2025 – July 2026

Digital Egypt Pioneers Initiative (DEPI)

- Selected for the competitive DEPI Cybersecurity track; conducted network and web application security assessments aligned with Mastercard's enterprise security framework
- Performed vulnerability identification, risk analysis, and security reporting within globally distributed security teams across multiple time zones
- Applied secure coding concepts and security testing methodologies throughout practical projects

HR Manager

December 2024 – Present

AITU DEV – Assiut, Egypt (Remote)

- Manage human resources for a tech development organization while maintaining active involvement in technical projects and development work

IT & Network Assistant

October 2022 – Present

Egyptian-German College of Technology

- Design and implement LAN infrastructure and secure Windows Server environments for laboratory network environments
- Support networking and Windows Server practical sessions and troubleshoot hardware, software, and connectivity issues
- Assist students with networking concepts, laboratory exercises, and network security protocols

Smart Home Research & Development Engineer

January 2025 – April 2025

Research & Development Team, Egyptian-German College of Technology

- Designed and developed a commercial-ready IoT-based Smart Home platform integrating embedded systems and a Flutter mobile application
- Led hardware selection, system architecture, PCB design, and automation scenarios using ESP8266/ESP32 and sensors
- Achieved 4th Place at the SCTE2025 International Conference on Technological Education, Royal Maxim Palace Kempinski, Cairo

Network Configuration Engineer (Freelance)

September 2020 – Present

Remote

- Design enterprise network infrastructures using Cisco technologies: VLANs, Inter-VLAN Routing, OSPF, BGP, NAT, ACLs, DHCP, DNS, HSRP, and Port Security
- Build enterprise network simulations in Cisco Packet Tracer and GNS3; prepare professional documentation, diagrams, and implementation reports

Front-End Developer (Freelance)

February 2023 – Present

Remote

- Develop responsive web interfaces using HTML5, CSS3, JavaScript, and Bootstrap, collaborating with clients on customized web solutions

Web Development Intern

May 2022 – September 2022

Information Technology Institute (ITI)

- Developed a complete Student Management System (PHP & MySQL) with administrator and student dashboards, registration, grading, and PDF reporting modules

GRADUATION PROJECT

IoT-Based Smart Village for Green Energy Optimization

Excellent Grade | Defended July 7, 2026

A fully integrated Smart Village ecosystem combining IoT, Edge Computing, Cloud Computing, Mobile Applications, and Embedded Systems into a unified platform for intelligent infrastructure management.

- Smart Home – automated lighting, gas/smoke/flame detection, RFID access control, remote monitoring
- Smart Irrigation – soil moisture, temperature, humidity, and rainfall-based automated water management
- Smart Parking & Traffic Management – real-time availability, reservation system, vehicle locator, adaptive signal control
- Smart Street Lighting, Waste Management & Surveillance – motion/daylight-based lighting, fill-level monitoring, ESP32-CAM live streaming
- Smart Power Management – intelligent energy distribution and monitoring

Technologies: ESP32, ESP8266, Flutter, Dart, C# .NET Web API, SQL Server, MQTT, REST APIs, PCB Design, Arduino IDE, Figma, Blender, AutoCAD, 3D Printing

PROJECTS

Smart Home System — 4th Place, SCTE2025 International Conference

- Commercial-ready smart home platform with gas leak detection & auto-shutoff, smart garage automation, solar energy monitoring, and Google Assistant voice control, monitored via a Flutter app

Hospital Enterprise Network Design

- Complete infrastructure design for a hospital (81 PCs, Cisco ISR 4321 router, 13 switches) featuring HSRP high availability, OSPF, VLANs, ACLs, NAT, and VoIP/HL7-DICOM integration

Enterprise Network Simulations

- Academic and freelance network designs featuring BGP filtering/redistribution, Private VLANs (PVLAN), QoS, and secure management via Syslog & SSH

Online Examination System — Excellent Grade

- Secure web-based examination platform (PHP/MySQL) with authentication, randomized question banks, and automated/instant grading

Student Management System

- Full-stack system for ITI with admin/student portals, grade reports, PDF export (DomPDF), and schedule management

Nand2Tetris Assembler

- Complete assembler built in C# (Parser, SymbolTable, Code classes) translating assembly language into machine code

EDUCATION

Bachelor of Information Technology

October 2022 – July 2026

Egyptian-German College of Technology – Assiut International Technology University

- Graduation Project: IoT-Based Smart Village for Green Energy Optimization — Excellent Grade

Technical Diploma in Technology

September 2019 – May 2022

Technological Secondary School, Assiut

- Graduation Project: Online Exam System (PHP, MySQL, HTML/CSS/JS) — Excellent Grade

TECHNICAL SKILLS

Networking: LAN/WAN Design, VLAN & Inter-VLAN Routing, Subnetting, OSPF, BGP, Static Routing, ACL, NAT, DHCP/DNS, HSRP, STP/VTP, Port Security, PVLAN, QoS, SDN Concepts, TCP/IP, Cisco Packet Tracer, GNS3, Windows Server

Cybersecurity: Vulnerability Assessment, Penetration Testing, Ethical Hacking Fundamentals, Secure Networking, Security Analysis & Reporting

Programming & Web: C#, Java, JavaScript, C++, HTML5, CSS3, Bootstrap, ASP.NET MVC, PHP, WordPress

IoT & Embedded / Mobile: ESP32, ESP8266, MQTT, REST APIs, Arduino, Sensor Integration, Flutter, Dart

Databases & Tools: Microsoft SQL Server, MySQL, Cisco Packet Tracer, GNS3, Visual Studio, VS Code, Git, Figma, Microsoft Office

Operating Systems: Windows, Windows Server, Linux (Red Hat basics)

Teaching & Educational: Curriculum Development, Lesson Planning, Technical Training, Student Assessment, Project-Based Learning, Educational Technology, Student Mentoring

CERTIFICATIONS

Cisco Networking Academy

- Cisco Certified Support Technician (CCST) – Cybersecurity (Score: 965/1000)
- CCNA: Enterprise Networking, Security & Automation (Aug 2025)
- CCNA: Switching, Routing & Wireless Essentials (Jul 2025)
- CCNA: Introduction to Networks (Jul 2025)
- Ethical Hacker, Cybersecurity Essentials, Networking Essentials, Introduction to Cybersecurity
- Introduction to IoT & Digital Transformation, Introduction to Data Science, English for IT 1 & 2

Digital Egypt Pioneers Initiative (DEPI)

- Cybersecurity Track — Specialization: Vulnerability Analyst & Penetration Tester

Information Technology Institute (ITI)

- Diploma in Web Development — Computer Networks, ASP.NET MVC, C# Programming, Agile Fundamentals

MaharaTech (ITI MOOCs)

- Red Hat System Administration I, Ethical Hacking, Cloud & Virtualization Concepts
- Network Virtualization Concepts, Python Basics, Computer Network Fundamentals
- Mobile Application Development, UX Design Fundamentals, PHP & MySQL Development

Additional Certifications

- Optical Fiber Systems – Telecom Egypt
- IT Technician – German Dual Training System (AHK Egypt)
- DEY – Creativa Innovation Hubs & NTI
- MySQL Database Management System

CONFERENCES & ACHIEVEMENTS

- 4th Place – Smart Home Project, SCTE2025 International Conference on Technological Education
- Research & Development Team Member, Egyptian-German College of Technology
- Participant, Practicing Software Leaders of Africa
- Participant, Innoventure 2026
- Participant, Youth Week – Assiut International Technological University

LANGUAGES

Arabic: Native

English: Good (Written & Spoken)